

SWE2 Level Design Document

Over-The-Air (OTA) Software Update System

Document Version: 2.0

Prepared By: SWE2 Engineer

Project Type: Automotive OTA System

Purpose: Detailed design document for RAG project and engineering reference

1. Introduction

1.1 Purpose

This document provides a Software Engineer 2 (SWE2) level design of an Automotive Over-The-Air (OTA) update system. It describes system architecture, module design, data flow, APIs, failure handling, and security considerations.

This document is intended for:

- SWE2 engineers
- OTA developers
- System designers
- RAG knowledge base

1.2 Scope

This document covers:

- System architecture
 - Component level design
 - Data flow
 - State machines
 - APIs
 - Error handling
 - Security
 - Deployment
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2. System Architecture

2.1 High Level Architecture

The OTA system consists of the following components:

- OTA Cloud Server
- OTA Gateway / TCU
- Update Agent
- Target ECU
- Bootloader
- Storage Manager

2.2 Architecture Layers

Cloud Layer

- Campaign Manager
- Update Manager
- Device Manager
- Authentication Service

Vehicle Layer

- OTA Client
- Download Manager
- Update Manager
- Verification Manager
- Install Manager

ECU Layer

- Update Agent
- Bootloader

3. Component Design

3.1 OTA Client

Responsibilities

- Check for updates

- Download updates
- Validate updates
- Install updates
- Report status

Sub Modules

- Update Checker
 - Download Manager
 - Verification Manager
 - Install Manager
 - Report Manager
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3.2 Update Checker

Responsibilities

- Periodic update check
- Manual trigger
- Campaign validation

Flow

1. Send device info
2. Receive manifest
3. Validate manifest
4. Trigger download

Data Structure

DeviceInfo

```
{  
    vehicle_id  
    ecu_id  
    software_version  
    hardware_version
```

}

3.3 Download Manager

Responsibilities

- Download firmware
- Resume download
- Parallel download

Features

- Chunk download
- Resume support
- Timeout handling

Download Flow

1. Request chunk
2. Receive chunk
3. Store chunk
4. Verify chunk
5. Continue download

3.4 Verification Manager

Responsibilities

- Check checksum
- Verify signature
- Validate manifest

Verification Steps

1. Validate checksum
 2. Validate signature
 3. Validate version
-

3.5 Install Manager

Responsibilities

- Send firmware to ECU
 - Monitor installation
 - Handle rollback
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4. State Machine

OTA Client State Machine

States:

- Idle
- Checking
- Downloading
- Verifying
- Installing
- Completed
- Failed

State Transitions:

Idle -> Checking

Checking -> Downloading

Downloading -> Verifying

Verifying -> Installing

Installing -> Completed

Installing -> Failed

5. Data Flow

OTA Update Flow

1. Vehicle boots
2. OTA client initializes
3. Check update

4. Download package
 5. Verify package
 6. Install package
 7. Report result
-

6. API Design

6.1 Check Update API

POST /check_update

Request:

```
{  
  "vehicle_id": "123",  
  "ecu": "ADAS",  
  "version": "1.0.0"  
}
```

Response:

```
{  
  "update_available": true,  
  "version": "1.1.0",  
  "url": "download_url",  
  "checksum": "abc123"  
}
```

6.2 Download API

GET /download

6.3 Report API

POST /report

7. Storage Design

Storage Locations

- Temporary storage
- Verified storage
- Backup storage

Storage Flow

Download -> Temporary

Verify -> Verified

Install -> Backup

8. Error Handling

Error Types

- Network error
- Verification failure
- Install failure
- Power failure

Retry Strategy

- Retry download
- Resume download
- Rollback firmware

9. Security Design

Security Features

- TLS communication
- Certificate validation
- Firmware signing
- Secure boot

Security Flow

1. Authenticate server

2. Download signed firmware
 3. Verify signature
 4. Install firmware
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10. Rollback Design

Rollback Conditions:

- Install failure
- Power loss
- Verification failure

Rollback Flow:

1. Bootloader detects failure
 2. Restore previous firmware
 3. Report rollback
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11. Logging

Log Types

- Update logs
- Error logs
- Installation logs

Example:

[INFO] Checking update

[INFO] Download started

[INFO] Download completed

12. Performance Considerations

- Background download
- Delta updates
- Parallel ECU updates

13. Deployment Design

Deployment Flow

1. Upload firmware
2. Create campaign
3. Assign vehicles
4. Trigger update

14. Folder Structure

ota

|-- client

|-- server

|-- ecu

|-- bootloader

|-- docs

15. Design Assumptions

- Reliable network
- Sufficient storage
- Secure boot supported

16. Future Enhancements

- Multi ECU update
- Delta updates
- Scheduling updates

17. RAG Keywords

OTA
Automotive
Firmware Update
Update Manager
Download Manager
Verification
Install Manager
Rollback
Security

End of Document